

Akash Sonth

Boston, MA, USA | akashsonth@gmail.com | +1 540-355-6787 | akashsonth.github.io

EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech) Blacksburg, VA, USA
M.S., Computer Engineering (concentration: Machine Learning), GPA: 3.88/4 Aug 2021 - Jul 2023

- Thesis: [Enhancing road safety through machine learning for prediction of unsafe driving behaviors](#)

Birla Institute of Technology and Science, Pilani (BITS Pilani) Pilani, India
B.E. (Hons.), Electrical and Electronics Engineering, GPA: 8.39/10 Aug 2015 - Jul 2019

- Thesis (at Brno University of Technology): [Vehicle density prediction and counting from monocular images](#)
- Member, SAE Formula Student team: End-to-end design and integration of a Formula Student race car.

TECHNICAL SKILLS

Programming	Python, C#, MATLAB, SQL, $\text{\LaTeX}$
Frameworks	PyTorch, PyG, Keras, TensorFlow, OpenCV, Open3D, scikit-learn
Robotics & Simulation	ROS, OpenStreetMap (OSM), Lanelet2, Unity, Raspberry Pi
Systems & Cloud	Git, Docker, Linux, Bash, CI/CD, AWS (EC2, S3)
Agentic AI	Microsoft Agent Framework

EXPERIENCE

Emerson (formerly **Aspen Technology**) Bedford, MA, USA
Data Scientist Aug 2023 - Present

- Developed an agentic framework for outlier detection, data quality assessment, and failure-relevant sensor selection in industrial time-series datasets, reducing analysis time from 2 hrs to under 5 mins per asset. (*U.S. Patent filed*)
- Designed an unsupervised equipment monitoring algorithm fusing expert-defined and generative-model-derived signatures, improving detection lead time by up to 10 hrs over production methods. (*U.S. Patent filed*)

Virginia Tech Transportation Institute Blacksburg, VA, USA
Graduate Research Assistant Dec 2021 - Aug 2023

- Designed a multi-stage perception pipeline to represent complex traffic scenes as semantic scene graphs, combining object detection, multi-object tracking, and Lanelet2-based road topology.
- Developed TransformerConv and GINEConv based graph neural networks for intersection safety prediction, achieving 79% accuracy on low-resolution (320×240 , 15 fps) videos.
- Adapted the Video Swin Transformer for explainable driver activity recognition using a visual dictionary of human actions, achieving 80.2% Top-1 accuracy on 10k+ naturalistic driving videos.
- Built a containerized Model Zoo for deployment of eye gaze estimation, multi-object tracking, and panoptic segmentation models.

Amazon Web Services Arlington, VA, USA
Software Development Engineer Intern May 2022 - Aug 2022

- Developed RESTful APIs for an enterprise email security analysis application, integrating Splunk logs with a Flask + Bootstrap interface to investigate spam, spoofing, and malware-related email activity and generate email reports.

Visual Computing Lab, Indian Institute of Science Bengaluru, India
Research Assistant Feb 2020 - Apr 2021

- Developed a two-stage self-supervised framework for person re-identification using pose-transformed training data, a ResNet-50 backbone, and a novel radial distance loss, achieving state-of-the-art 93.6% Rank-1 accuracy on Market-1501 (32k+ images).
- Implemented distributed multi-node, multi-GPU training using the NCCL backend, enabling 4x larger batch-sizes.

- Investigated respiratory disease detection from cough sounds using data from 20 patients collected in collaboration with hospitals in the New Delhi area, comparing sequential audio and spectrogram-based representations.
- Developed a classification pipeline using transfer learning with CNNs (AlexNet, VGG19) for feature extraction followed by Support Vector Machine classification, achieving 78% accuracy on respiratory disease classification.

TEACHING EXPERIENCE

BITS Pilani | F312 Neural Networks and Fuzzy Logic - Teaching Assistant

Jan 2019 - May 2019

- Assisted with course instruction and coursework, and mentored undergraduate students on course projects.

PUBLICATIONS [[Google Scholar](#)]

- **Akash Sonth**, Lynn Abbott, Abhijit Sarkar. “Intersection Safety Modeling Using Semantic Scene Graph and Graph Neural Network.” IEEE Intelligent Vehicles Symposium (IV), 2025. [[Paper](#)][[Project Page](#)]
- **Akash Sonth**, Abhijit Sarkar, Hirva Bhagat, Lynn Abbott. “Explainable Driver Activity Recognition Using Video Transformer in Highly Automated Vehicle.” IEEE Intelligent Vehicles Symposium (IV), 2023. [[Paper](#)][[Project Page](#)]
- **Akash Sonth**, Harshavardhan Settibhaktini, Ankush Jahagirdar. “Vehicle speed determination and license plate localization from monocular video streams.” International Conference on Computer Vision & Image Processing (CVIP), 2018. [[Paper](#)]
- Siddharth Seth, **Akash Sonth**, Anirban Chakraborty. “Pose-Transformation and Radial Distance Clustering for Unsupervised Person Re-identification.” British Machine Vision Conference (BMVC), 2021. [[Paper](#)][[Project Page](#)]
- Hirva Bhagat, Sandesh Jain, Lynn Abbott, **Akash Sonth**, Abhijit Sarkar. “Driver Gaze Fixation and Pattern Analysis in Safety Critical Events.” IEEE Intelligent Vehicles Symposium (IV), 2023. [[Paper](#)]

AWARDS & FELLOWSHIPS

- Oral Presentation, IEEE Intelligent Vehicles Symposium (IV) (Top 7% of accepted papers), 2025
- IEEE Intelligent Transportation Systems Society (ITSS) flagship conference travel grant, 2025
- Google Research India Summer School (Computer Vision track), 2020 - one of 50 students selected nationwide.
- Virginia Tech, Center for Human-Computer Interaction (CHCI) conference award, 2023
- Virginia Tech, Graduate and Professional Student Senate (GPSS) conference award, 2023

SELECTED TALKS

- “Intersection Safety Modeling Using Semantic Scene Graph and Graph Neural Network.” Invited talk at 2nd Workshop on Secure connected vehicles: Digital Twin, UAVs, and Smart Transportation, IEEE Intelligent Vehicles Symposium (IV), 2025.
- “Real-Time Risk Prediction at Signalized Intersection Using Graph Neural Network.” Virginia Tech Transportation Institute (VTTI), Safe-D webinar, 2023 [[Presentation](#)]